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***Northern Plating Company Site  
Cadillac, Michigan  
Second Five Year Review***

**I. Introduction**

EPA Region 5 has conducted a five year review of the remedial actions implemented at the Northern Plating Company Site in Cadillac, Michigan. This five year review represents the second review for the Northern Plating Site. This review was conducted from May 15, 2000 to July 19, 2000 and a site visit was conducted on July 19, 2000. This site visit was attended by Dion Novak, Remedial Project Manager, Larry Campbell-City of Cadillac Utilities Department, Pete Dauks, Tetra Tech - City of Cadillac contractor, and Jim Skipper-Michigan Department of Environmental Quality. This report documents the results of this review. The purpose of five year reviews is to determine whether the remedy at the Site remains protective of human health and the environment. The methods, findings, and conclusions of these reviews are documented in five year review reports. In addition, five year review reports identify deficiencies found during the review, if any, and identify recommendations to address them.

This review is required by statute and is a Level 1 review. EPA must implement five year reviews consistent with the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) and the National Oil and Hazardous Substances Pollution Contingency Plan (NCP). CERCLA Section 121 (c) as amended states:

If the President selects a remedial action that results in any hazardous substances, pollutants or contaminants remaining at the Site, the President shall review such remedial action no less often than each five years after initiation of such remedial actions to assure that human health and the environment are being protected by the remedial action being implemented.

The NCP Part 300.430 (f)(ii) of the Code of Federal Regulations (CFR) states:

If a remedial action is selected that results in hazardous substances, pollutants or contaminants remaining at the Site above levels that allow for unlimited use and unrestricted exposure, the lead agency shall review such action no less often than every five years after the initiation of the remedial action.

This is the second five year review conducted for the Northern Plating Site. The triggering action for this statutory review is the date of actual on-Site remedial action construction, which was October 4, 1988. Because hazardous substances remain at the Site above levels that allow for unlimited use and unrestricted exposure, a five year review is required by statute.

## **II Site Chronology**

Table 1 below lists the chronology of events for the Northernaire Plating Site.

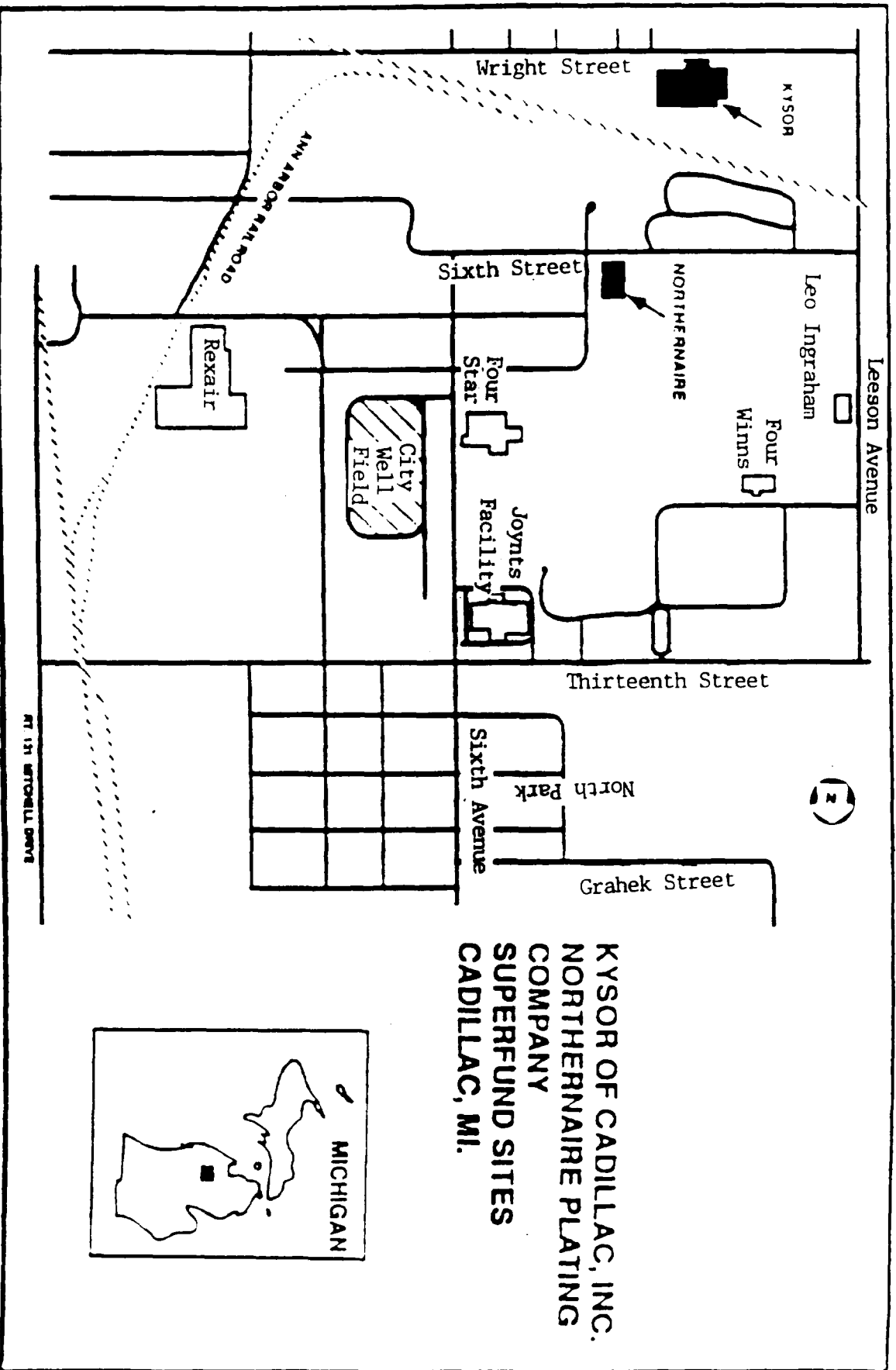
**Table 1**

| <b><u>Date</u></b> | <b><u>Event</u></b>                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------|
| 1971               | Northernaire electroplating operations began                                                               |
| 5/81               | Northernaire ceases operations                                                                             |
| 7/19/82            | Michigan Department of Natural Resources (MDNR) inspection of Northernaire Site                            |
| 9/7/82 - 2/2/83    | Preliminary Assessment by EPA FIT contractor                                                               |
| 3/83               | Preliminary Site inspection by EPA                                                                         |
| 6/28/83            | EPA determines immediate removal is necessary                                                              |
| 7/5/83 - 8/3/83    | EPA removal action at Northernaire Site                                                                    |
| 9/8/83             | Northernaire Site finalized on NPL                                                                         |
| 7/85               | Feasibility study completed for Site                                                                       |
| 9/11/85            | Record of Decision (ROD) for source control(operable unit 1) including soil, sewer and concrete excavation |
| 1/87               | Supplemental RI completed for chromium plume                                                               |
| 8/88               | Cadillac area contamination Remedial Investigation (RI) completed                                          |
| 9/89               | Cadillac area contamination Feasibility Study (FS) completed                                               |
| 9/29/89            | ROD signed for Kysor/Northernaire co-mingled groundwater contamination                                     |
| 5/16/90            | Unilateral Administrative Order (UAO) issued for Remedial Design                                           |
| 4/92               | MDNR issues Remedial Action Summary Report                                                                 |
| 1/30/95            | UAO issued for Remedial Action                                                                             |
| 3/6/95             | Final design for area groundwater remedy approved by EPA                                                   |

## **III Background**

The Site is located at 1002 Sixth Street in the Cadillac Industrial Park, in an area of approximately one mile square within the City of Cadillac. It is bordered by industrial, municipal, and undeveloped property. The Site was listed on the National Priorities List in July, 1982. In 1983, a removal action was conducted to remove drums and a sewer line from the Site (See Figure 1). Subsequently, a remedial investigation/feasibility study (RI/FS) was conducted by the Michigan Department of Natural Resources (MDNR), which determined that remedial actions were needed to protect public health and the environment from contact with contaminated soils (operable unit 1). A Record of Decision (ROD) was signed for operable unit 1 on September 11, 1985 (See Figure 1).

Figure 1



An additional RI/FS was conducted on the area wide groundwater contamination, which determined that remedial actions were also necessary for contaminated groundwater (operable unit 2). The RI/FS determined that the groundwater contamination associated with the Northernnaire Site was co-mingled with groundwater contamination associated with the Kysor Industrial Corporation Superfund Site. A ROD was signed for operable unit 2 of the Northernnaire Site on September 29, 1989. This ROD required the groundwater portion of the Northernnaire Site to be addressed in tandem with the groundwater contamination associated with the Kysor Site.

The remedial design and remedial action for operable unit 1 of the Northernnaire Site were performed by a MDNR contractor under a cooperative agreement with the U.S. EPA.

The remedial design for operable unit 2 of the Northernnaire Site and the entire Kysor Site was performed by a PRP in response to a unilateral administrative order (UAO) issued on May 16, 1990. Remedial action for operable unit 2 of the Northernnaire Site and the entire Kysor Site was performed by PRPs in response to two additional UAOs issued on January 30, 1995. A third UAO was issued on April 11, 1995 to two additional PRPs. A Local Development Finance Authority (LDFA) associated with the City of Cadillac financed construction and is operating the extraction system.

#### **IV Remedial Actions**

- |                    |                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/4/88 - 11/17/88 | Approximately 1016 cubic yards of soil were excavated from the Site, including areas to the west of and inside of the Northernnaire building. The process sewer line inside of the building and private sewer line serving the building was also excavated.                                                                                                          |
| 1/11/91, 5/8/91    | Pre-final and final inspections for operable unit 1.                                                                                                                                                                                                                                                                                                                 |
| 9/29/95 - 8/21/96  | Construction of groundwater Site remedy (operable unit 2 combined with the Kysor Site groundwater cleanup) which includes an 18 well extraction system with collection and discharge piping, a groundwater treatment system utilizing carbon adsorption to treat hexavalent chromium and air stripping to treat VOCs with surface water discharge to the Clam River. |

#### **Remedy Implementation**

The design for the site remedial actions was completed in March, 1995. Construction of the site remedial actions began in September, 1995 and was completed in August, 1996.

Static water level data since 1996 have indicated that the groundwater plume has been adequately contained with the 18 well extraction system. This is based on water level monitoring, which occurred on a quarterly basis for the first year of operation and on an annual basis since late 1997. Contaminant concentration data for VOCs and hexavalent chromium have consistently decreased since startup of the extraction system. In July 1997, the granular activated carbon contactor was taken off-line because hexavalent chromium had broken through the contactors. This problem was corrected and the contactor was placed back in service in December 1997. Since that time, the effluent from the treatment system has not exceeded discharge standards.

### **System Operations/O&M**

The groundwater extraction system has operated continuously since startup with no problems encountered. All of the extraction wells have operated continuously since being placed on-line. Weekly samples are collected of extraction system influent and effluent and analyzed for verification of compliance with system discharge limits.

### **Progress Since the Last Five Year Review**

This is the second five year review for the Site. The last review, dated September 1995, recommended that the groundwater extraction system be constructed and operated which is documented in this review.

### **V Five Year Review Findings**

Off-Site disposal of soils excavated from near the Northernair building and concrete excavated from inside the Northernair building was completed in 1991, after compliance issues with the RCRA Land Disposal Restrictions and associated new treatment requirements were resolved.

The groundwater cleanup is in place and has been operational since August 1996. The remedial goals for the Kysor/Northernair cleanup were to construct a groundwater extraction and treatment system to remediate contaminated groundwater at the Site (See Figure 1). Groundwater treatment includes carbon adsorption for hexavalent chromium treatment and packed tower air stripping for VOC contamination.

***Figure 1 Groundwater cleanup standards***

| <u>Compound</u> | <u>Cleanup Standard</u> |
|-----------------|-------------------------|
| 1,1,1-TCA       | 200 ppb                 |
| trans 1,2-DCA   | 70 ppb                  |
| 1,1-DCE         | 5 ppb                   |
| 1,2-DCA         | 5 ppb                   |

|                    |         |
|--------------------|---------|
| methylene chloride | 5 ppb   |
| PCE                | 1 ppb   |
| TCE                | 5 ppb   |
| xylenes            | 440 ppb |
| toluene            | 40 ppb  |

***Figure 2 Effluent criteria for discharge to the Clam River***

| <u>Compound</u>       | <u>Discharge Criteria</u> |
|-----------------------|---------------------------|
| 1,2 - dichloroethane  | 5 ppb                     |
| 1,1,1-trichloroethane | 5 ppb                     |
| trichloroethene       | 5 ppb                     |
| tetrachloroethane     | 5 ppb                     |
| hexavalent chromium   | 8.3 ppb                   |
| total chromium        | 59 ppb                    |

***Figure 3 Soil cleanup criteria***

| <u>Compound</u>       | <u>cleanup criteria</u> |
|-----------------------|-------------------------|
| trichloroethylene     | 0.07 ppb                |
| xylene                | 141 ppb                 |
| 1,1,1 trichloroethane | 7.6 ppb                 |
| toluene               | 724 ppb                 |

Soil cleanup levels were calculated so that residual contamination in the soil would not result in a continuing source of contamination to the groundwater above groundwater cleanup standards (See Figure 3).

The annual report for 1997 recommended that, because the extraction system was achieving performance objectives, the groundwater monitoring frequency be changed from quarterly to annually, which was determined to be sufficient to monitor the status of groundwater in the Site area. This annual monitoring began with the October 1998 sampling event for the compliance monitoring wells. A select group of monitoring wells designated by the EPA are still monitored quarterly.

Static water level data since 1996 have indicated that the groundwater plume has been adequately contained with the 18 well extraction system. This is based on water level monitoring, which occurred on a quarterly basis for the first year of operation and on an annual basis since late 1997. Contaminant concentration data for VOCs and hexavalent chromium have consistently decreased since startup of the extraction system. In July 1997, the granular activated carbon contractor was taken off-line because hexavalent chromium had broken through the contactors. This problem was corrected and the contractor was placed back in service in December 1997. Since that time, the effluent from the treatment system has not exceeded discharge standards.

A request was received from the City of Cadillac that requested a modification to the September 1989 Record of Decision for the cleanup standard for tetrachloroethene (PCE) in February 1999. The City of Cadillac requested that the cleanup standard for PCE for the Site be changed from 1 ppb to 5 ppb, to be consistent with the standard that the Michigan Department of Environmental Quality (MDEQ) was using on another cleanup Site.

## **VI Assessment**

The groundwater extraction system continues to provide containment of the contaminated groundwater plume, as evidenced by water level data that is continuing to be collected. This containment continues to be documented by groundwater data presented in annual performance monitoring reports. As mentioned above, VOC concentrations have generally declined in the shallow aquifer and have remained relatively constant in the intermediate aquifer. Hexavalent chromium concentrations have declined to below cleanup criteria in the latest sampling event (See Figure 1).

Groundwater treatment has continued to reduce the amount of VOC and hexavalent chromium contamination in the underlying aquifer. Influent concentrations to the treatment facility have decreased as a result of operations indicating that the system is meeting design standards. The treatment system has consistently achieved compliance with NPDES effluent limits (See Figure 2) for discharge to the Clam River.

## **VII Deficiencies**

At present, the Site remedy is performing adequately and achieving ROD performance standards. There are no deficiencies at present with any component of the implemented remedy.

## **VIII Recommendations and Required Actions**

This five year review report has summarized the current remedial activities at the Site and also describes the future monitoring activities at the Site. The following actions are recommended for continued operation and maintenance of Site remedies.

In conjunction with the Kysor cleanup, the groundwater extraction and treatment system should continue to be operated, providing downgradient containment of the VOC and hexavalent chromium contaminant plumes. It is recommended that the current groundwater monitoring data be studied to prepare a trend analysis report to evaluate contaminant trends in groundwater near the Site. This report should also include an analysis of the current monitoring network to determine the optimal configuration and frequency for Site remedy monitoring. The results of this analysis can be used to evaluate the performance of the extraction system at reducing

contaminant levels and also be used to evaluate potential contaminant contributions from neighboring sites.

It is also recommended that the groundwater extraction and treatment system be studied to determine if it can undergo optimization to improve performance while maintaining its protectiveness.

It is recommended that close coordination be established with the ongoing studies of neighboring sources of potential groundwater contamination, such as the Mitchell Site, the Rexair Site and the Paulstra CRC Sites with the Michigan Department of Environmental Quality so that any adverse impacts on the Kysor/Northernair system can be quantified and prevented.

It is also recommended that the February 1999 proposed change in the cleanup standard for PCE be assessed and resolved between the EPA and the MDEQ, with results communicated to the City of Cadillac.

#### **IX      Protectiveness of Remedy**

The remedy for the Northernair Plating Site is protective of human health and the environment.

#### **X      Next Review**

The next five year review will be conducted within five years of the completion of this report, which will be July 2005.

  
for William E. Muno, Director  
Superfund Division  
U.S. EPA Region 5

July 26, 2000  
Date